GELDIUM | MODEL RISK & ANALYTICS

Delinquency Prediction Model Development & Validation Report

Empirical assessment of five classification algorithms, operational targeting, explainability, calibration, and exploratory fairness.

ANALYSIS	SCOPE
Dataset	500 customer records; 80 delinquent (16%)
Holdout evaluation	125 customers; 20 delinquent and 105 non-delinquent
Algorithms	Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, Neural Network (MLP)
Reference	Prior-probability baseline (16%)
Assessment	Exploratory research; not approved for production credit decisions

Prepared 11 October 2026 | Evidence: supplied modeling outputs, tables, and ten uploaded figures.

Executive decision brief

Decision: DO NOT DEPLOY. Across the evaluated holdout, every trained model had ROC-AUC below 0.50. The prior-probability baseline produced the lowest Brier score (0.1344), and threshold-based alerts yielded many false positives or missed all delinquent customers. There is no reliable evidence of an actionable delinquency risk ranking in this experiment.

Decision-critical findings

Finding	Evidence	Implication
Low prevalence	80 / 500 customers (16%) are delinquent	Accuracy alone rewards predicting the majority class.
Model selection	Logistic Regression leads training-only cross-validation PR-AUC: 0.208 ± 0.037	Use it as a research benchmark, not as a validated operational model.
Holdout ranking	Random Forest has highest holdout AP in results CSV: 0.244; ROC-AUC: 0.400	Higher AP alone does not establish a deployable model.
Classification trade-off	Logistic Regression: 5 TP, 44 FP, 15 FN, 61 TN at 0.50	Only 25% of delinquent cases found; precision 10.2%.
Calibration	Baseline Brier 0.1344 vs Logistic Regression 0.2637	Model probability scores should not be presented as trustworthy default probabilities.
Fairness	Employment subgroup rates vary on small holdout subsets	Audit is exploratory, not proof of discrimination or fairness.

Evidence and consistency note

The supplied chart images and the saved Geldium_Model_Results.csv do not match in every metric. For example, the chart reports Random Forest holdout AP = 0.269 and Gradient Boosting AP = 0.159, whereas the CSV records 0.244 and 0.186 respectively. ROC chart labels also differ slightly from the CSV. This report uses the CSV as the numerical reference for tables and decisions, reproduces all ten supplied charts unchanged, and flags discrepancies. Re-run all notebook cells with fixed seeds and export synchronized figures and tables before formal submission or audit.

1 Data, target, and experimental design

The source dataset contains 500 customer observations with a binary Delinquent_Account outcome. The positive class comprises 80 records (16%). The holdout shown in the supplied figures contains 125 observations (20 positive, 105 negative), consistent with a 25% stratified test split. Training-only five-fold cross-validation was used to compare candidates before final holdout assessment.

Feature groups and data-quality considerations

Candidate predictors include credit utilization, credit score, debt-to-income ratio, income, loan balance, missed payments, account tenure, age, employment status, credit-card type, location, and monthly repayment statuses. The EDA identified missing income, loan balance and credit score values, as well as utilization above 100% in a few records. Such anomalies should be checked against source-system definitions; outliers must not be deleted solely because they are unusual.

The six monthly payment-status fields require a strict time-line audit. If any of those fields occur after the delinquency outcome being predicted, including them constitutes target leakage. A valid deployment design must specify an observation date, a future prediction horizon, and a feature availability cut-off.

Modeling controls

A defensible workflow fits imputation, scaling and encoding inside training folds only, uses stratification to preserve rare events, and reserves the holdout for a single final assessment. Hyperparameter tuning, threshold choice, calibration, and feature selection must use training/validation data—not the test labels. A temporal split is preferable if dates and prospective outcomes are available.

Metrics and operating definitions

Measure	Interpretation
Average precision / PR-AUC	Ranking quality for the rare delinquent class; prevalence 0.16 is a useful reference.
ROC-AUC	Probability that a randomly selected delinquent customer scores above a non-delinquent customer.
Recall	Share of actual delinquent customers flagged.
Precision	Share of flagged customers actually delinquent.
Brier score	Mean squared error of predicted probabilities; lower is better.
Lift / cumulative gains	Share of delinquencies captured when reviewing the highest-ranked customers.

2 Comparative model results

Model	CV AP ± SD	Test AP	Test ROC-AUC	Brier
Baseline	0.160 ± 0.000	0.160	0.500	0.134
Logistic Regression	0.208 ± 0.037	0.213	0.457	0.264
Decision Tree	0.182 ± 0.028	0.146	0.446	0.276
Random Forest	0.186 ± 0.037	0.244	0.400	0.176
Gradient Boosting	0.177 ± 0.036	0.186	0.411	0.143
Neural Network (MLP)	0.206 ± 0.038	0.182	0.443	0.159

Average precision (AP) and area under the precision–recall curve are treated as the same summary measure in the saved results. Although Random Forest leads holdout AP in the CSV, choosing it retrospectively based on test performance would invalidate the original model-selection procedure. Logistic Regression was the preselected cross-validation leader.

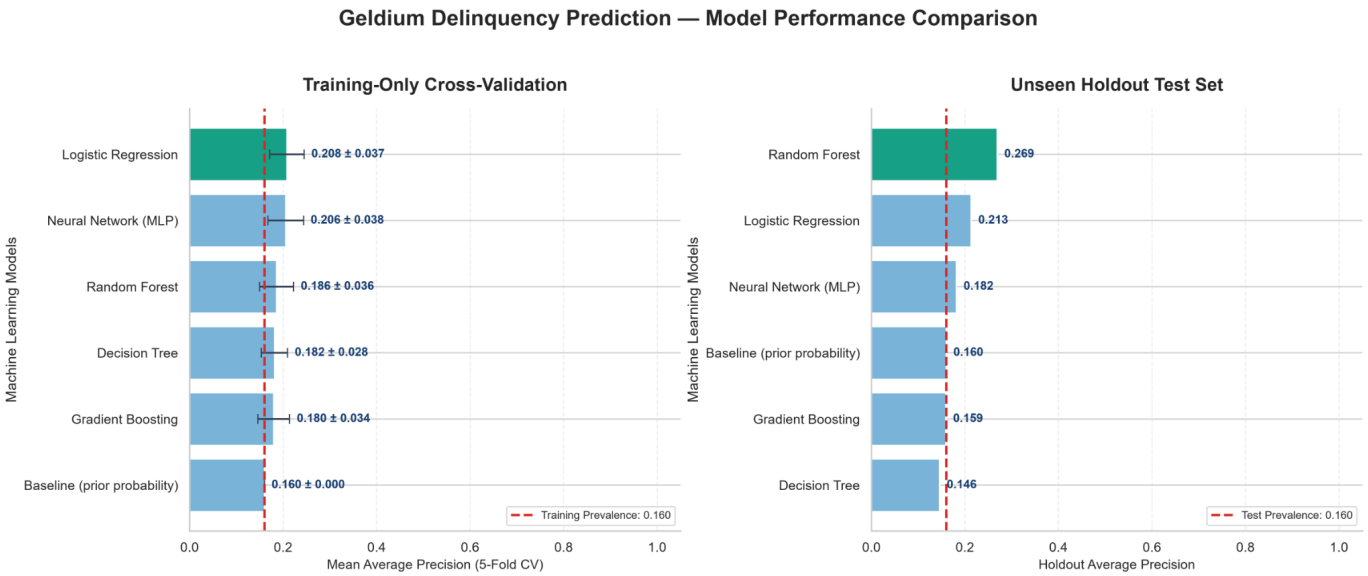


Figure 1. Cross-validation versus holdout average precision. Image labels reflect the supplied chart export; CSV values take precedence where inconsistent.

3 Discrimination: ROC analysis

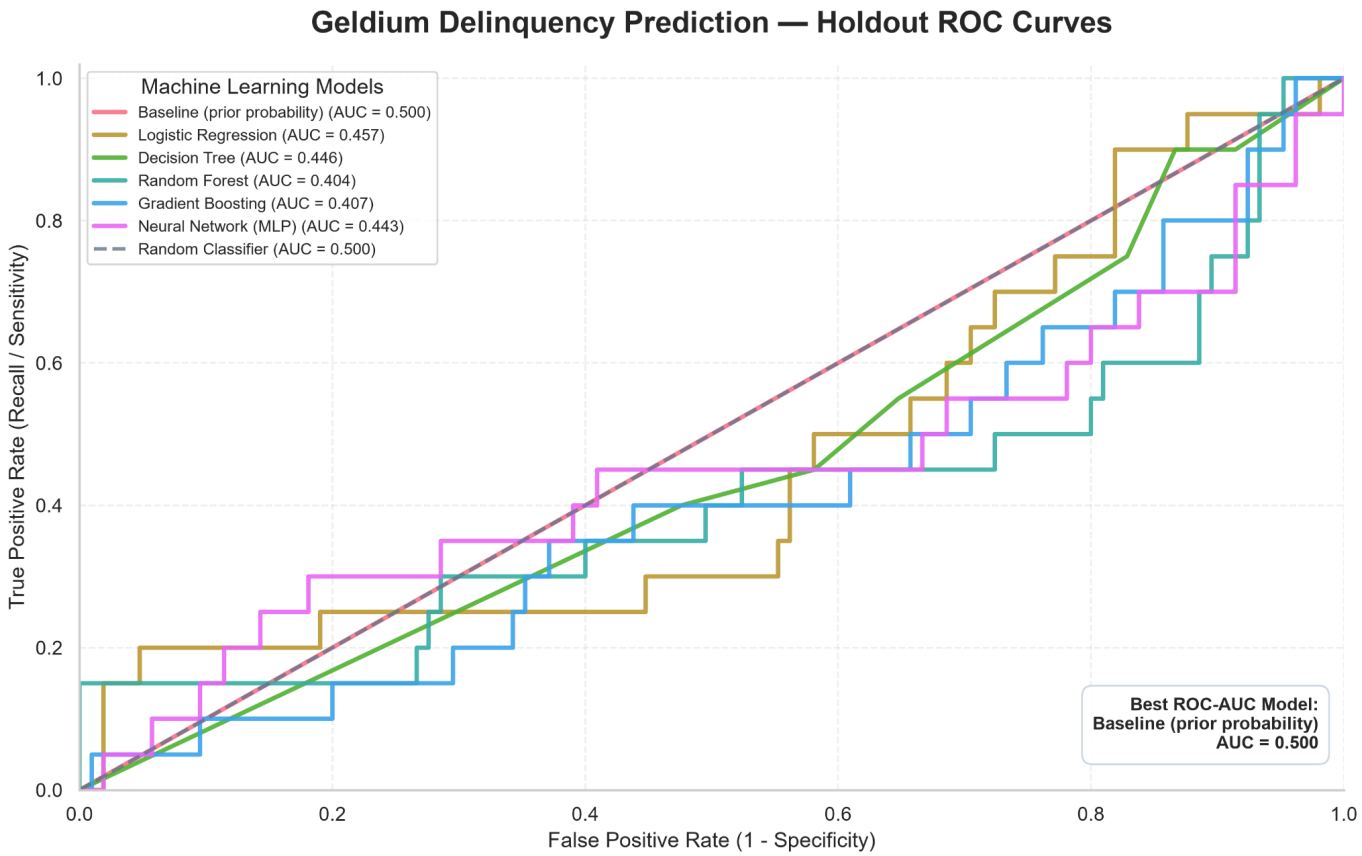


Figure 2. Holdout ROC curves for all candidates and the constant-risk baseline. The diagonal represents chance-level ranking.

All trained models fall below chance on this holdout (CSV ROC-AUC range 0.400–0.457). This does not establish that their learned relationships are universally reversed; with only 20 positive test cases, ranking estimates can be unstable. Nonetheless, the results do not support customer-level risk discrimination.

4 Precision–recall analysis

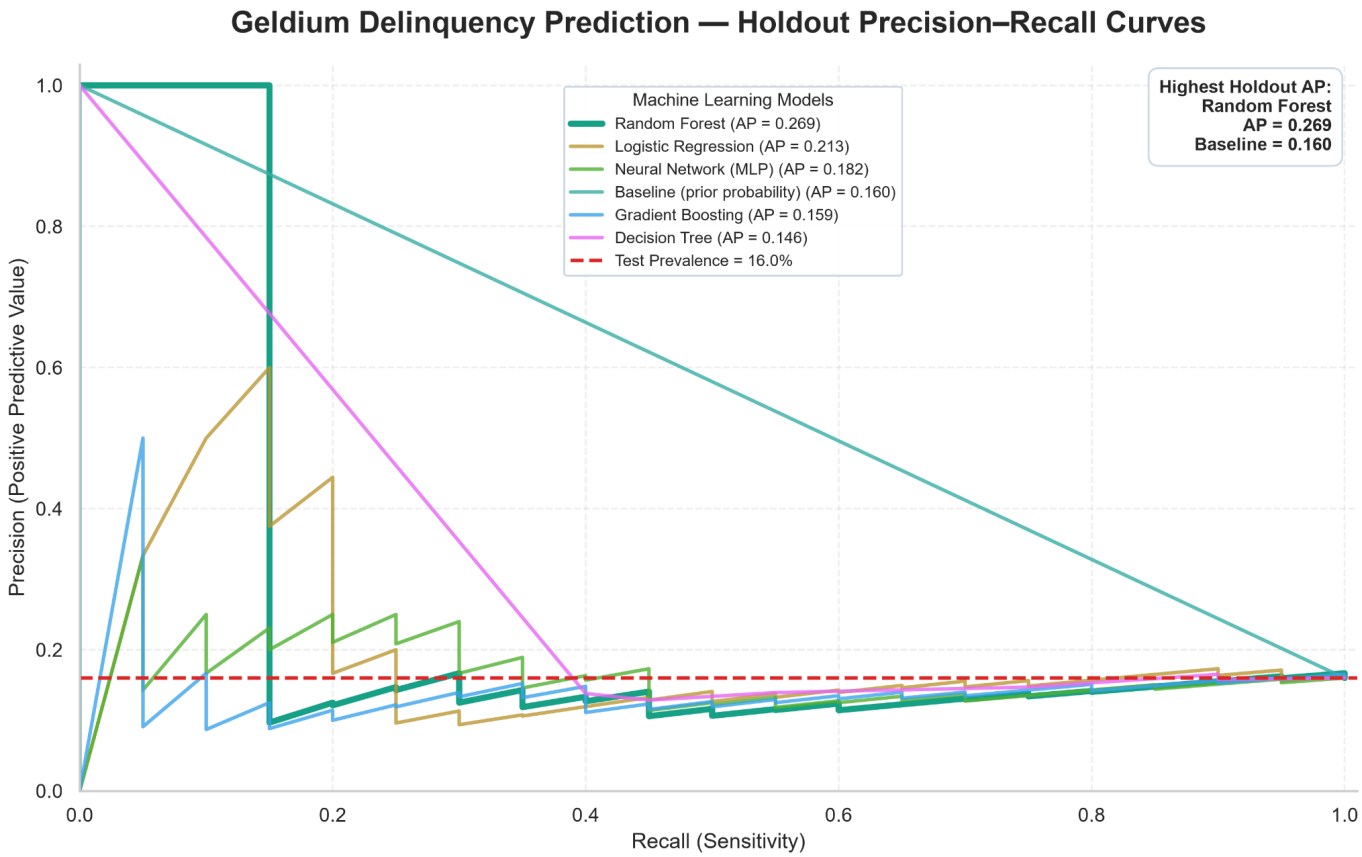


Figure 3. Holdout precision–recall curves. Early spikes can be caused by very few high-ranked observations.

The 16% positive prevalence is the reference AP for a constant-score baseline. CSV holdout AP values range from 0.146 to 0.244. A model can obtain AP above prevalence through limited pockets of correct rankings while still having ROC-AUC below 0.50. Neither metric alone should be used to justify deployment; evaluate both alongside calibration, subgroup reliability and operational review costs.

5 Classification errors at the 50% threshold

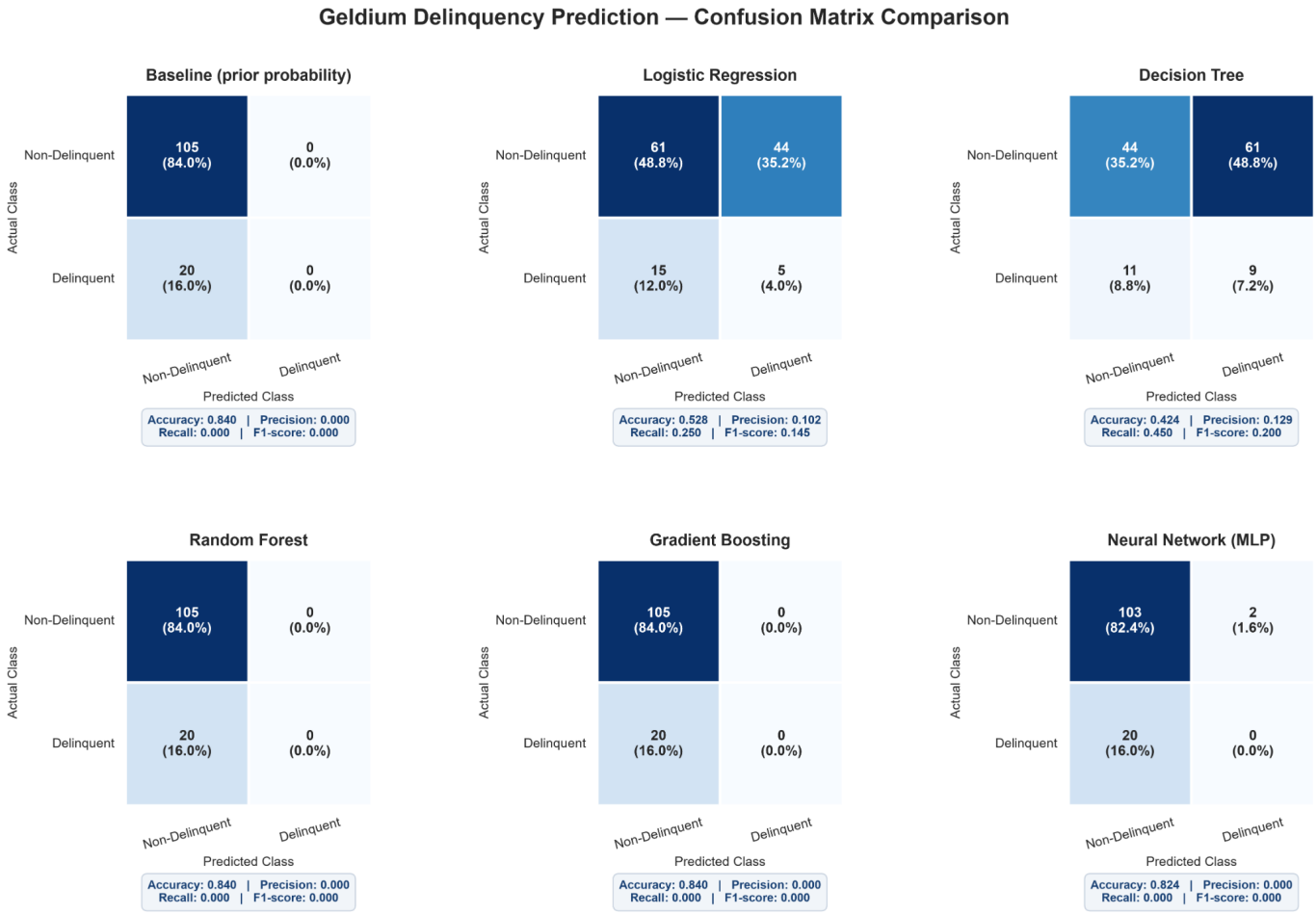


Figure 4. Holdout confusion matrices. Each matrix represents 125 customers; positive class = delinquent.

Operational confusion matrix table

Model	TP	FP	FN	TN	Precision	Recall
Baseline	0	0	20	105	0.0%	0.0%
Logistic Regression	5	44	15	61	10.2%	25.0%
Decision Tree	9	61	11	44	12.9%	45.0%
Random Forest	0	0	20	105	0.0%	0.0%
Gradient Boosting	0	0	20	105	0.0%	0.0%
Neural Network (MLP)	0	2	20	103	0.0%	0.0%

Logistic Regression identifies 5 of 20 delinquent customers but incorrectly flags 44 non-delinquent customers. Decision Tree identifies 9 delinquent customers with 61 false positives. Random Forest and Gradient Boosting flag no customers at the 50% threshold. The baseline achieves 84% accuracy only because it predicts the majority class for everyone; it catches no delinquent customers.

6 Risk-score distributions

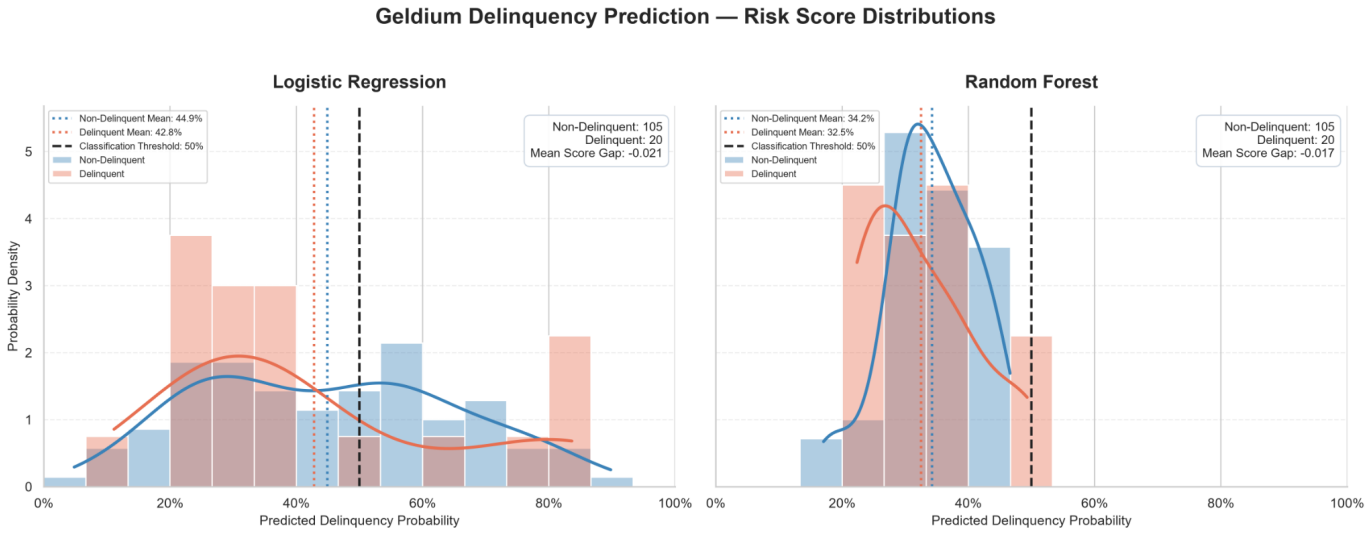


Figure 5. Predicted-score distributions for Logistic Regression and Random Forest, separated by actual delinquency status.

The average predicted score is slightly lower for delinquent customers than for non-delinquent customers in both displayed models: Logistic Regression 42.8% versus 44.9%, and Random Forest 32.5% versus 34.2%. The strong overlap and reversed mean-score gap reinforce the weak discrimination finding. The dotted 50% threshold is illustrative, not an optimized operating point.

7 Calibration and probability reliability

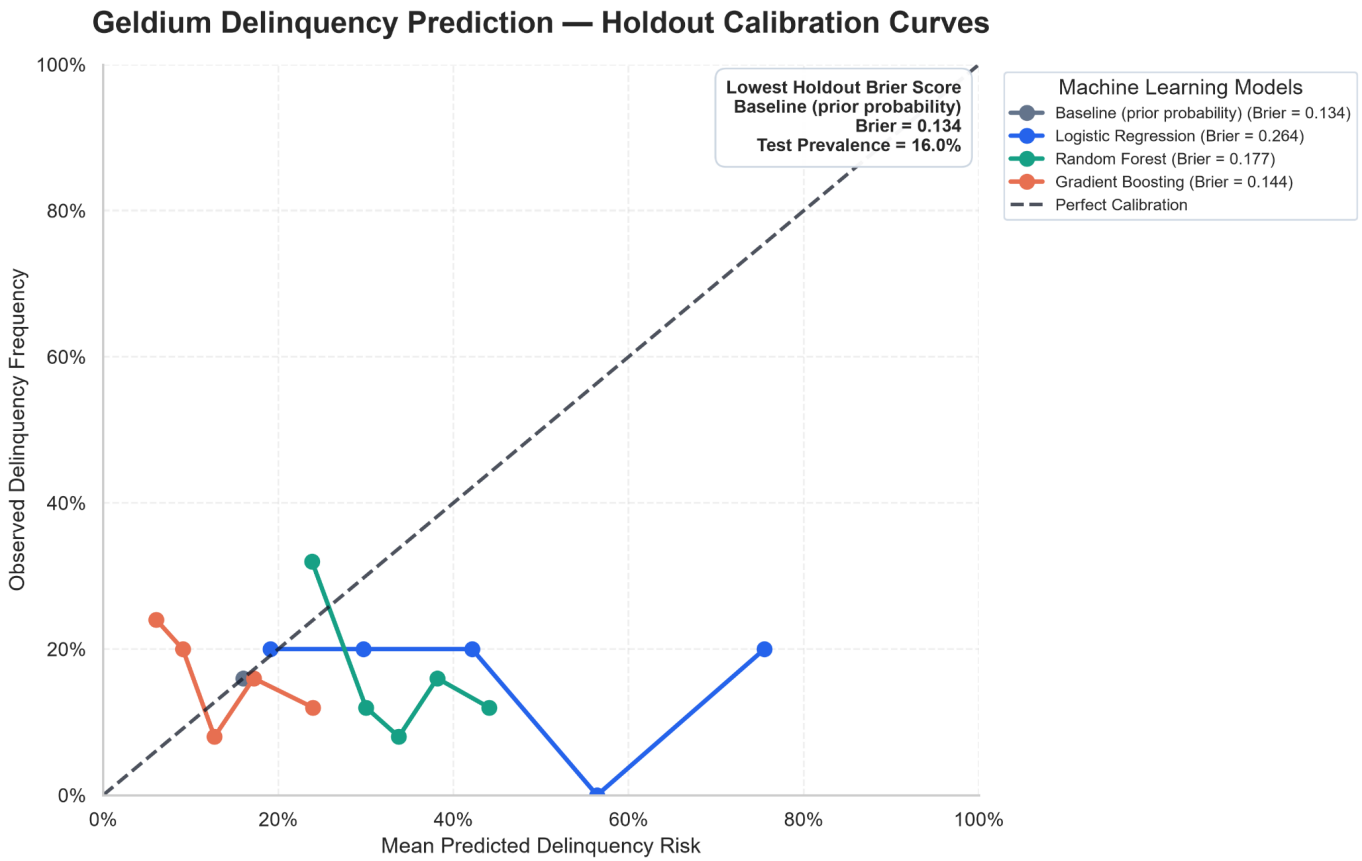


Figure 6. Observed delinquency frequency versus average predicted risk by score bin. Small-bin estimates may be unstable.

The constant 16% baseline has the lowest test Brier score (0.1344). Logistic Regression is markedly worse (0.2637), as are Decision Tree (0.2763), Random Forest (0.1757), Gradient Boosting (0.1433), and MLP (0.1587). The visible deviations from the diagonal indicate that raw model scores should not be interpreted as calibrated default probabilities. Calibration should be learned on a separate validation subset or via nested cross-validation, then tested only once on untouched data.

8 Decision threshold and review workload

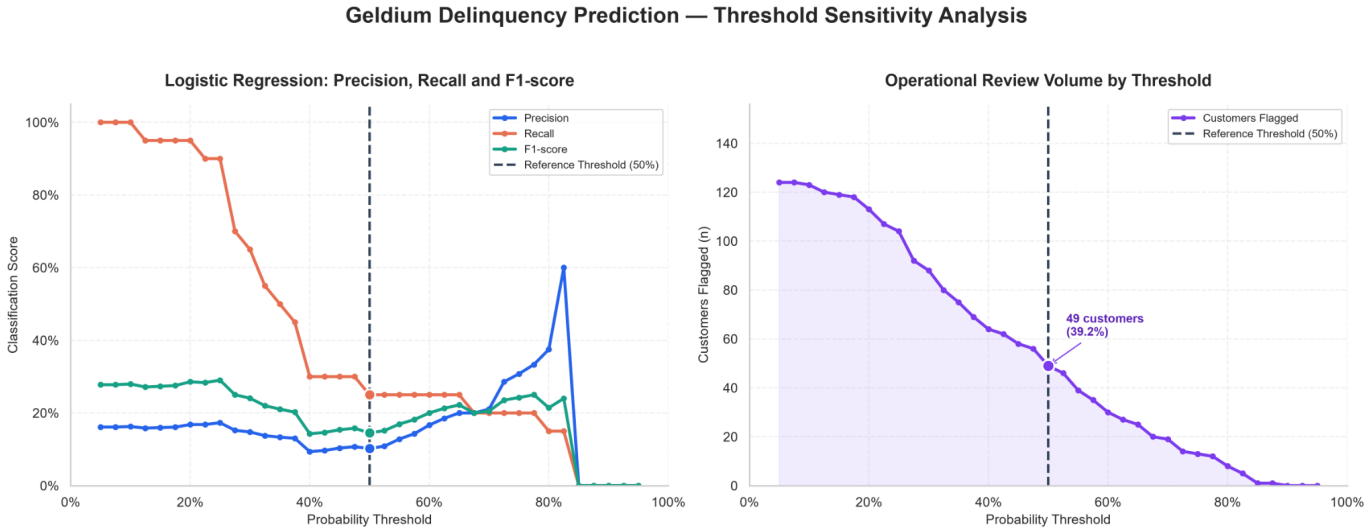


Figure 7. Logistic Regression precision, recall, F1 and flagged-customer volume across probability thresholds.

At the reference 50% cutoff, 49 of 125 customers (39.2%) are flagged, but only 5 are truly delinquent. Lowering the cutoff increases recall while driving review volume and false positives upward. Increasing the cutoff may reduce workload but misses more positive cases. The correct threshold requires an explicit intervention capacity and cost model (e.g., contact costs, expected recovery, customer harm, false-positive burden), chosen on validation data rather than the test set.

9 Cumulative gains and targeting effectiveness

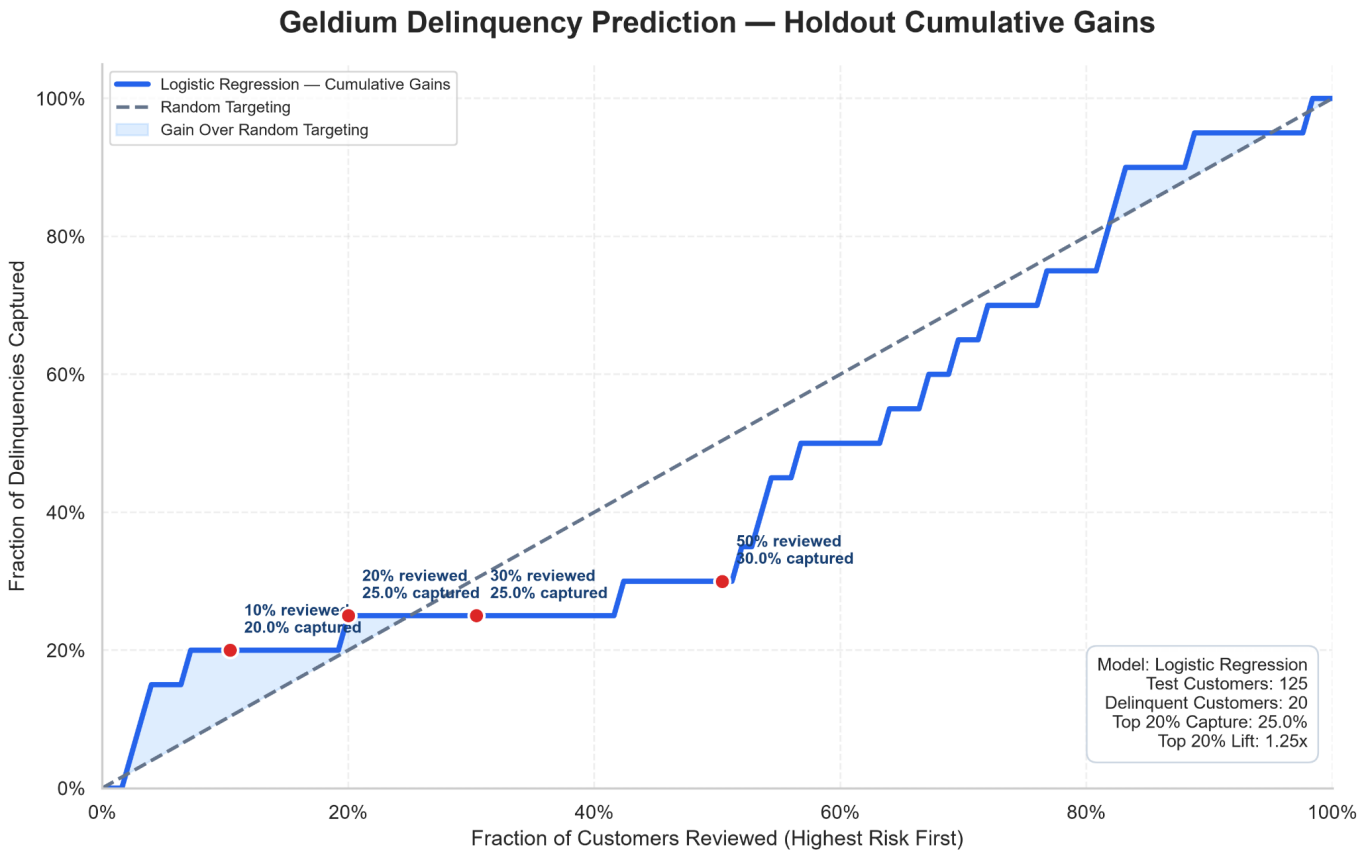


Figure 8. Fraction of observed delinquencies captured as the highest-scored customers are reviewed first.

For Logistic Regression, reviewing the highest-ranked 20% of holdout customers captures 25% of delinquencies (5 of 20): 1.25× lift over random targeting at that point. At 30% reviewed, capture remains 25%, below the 30% random expectation; at 50% reviewed, capture is only 30%, substantially below random. A single favorable top-decile or top-quintile point should not be treated as robust lift when the full curve is weak.

10 Explainability: permutation importance

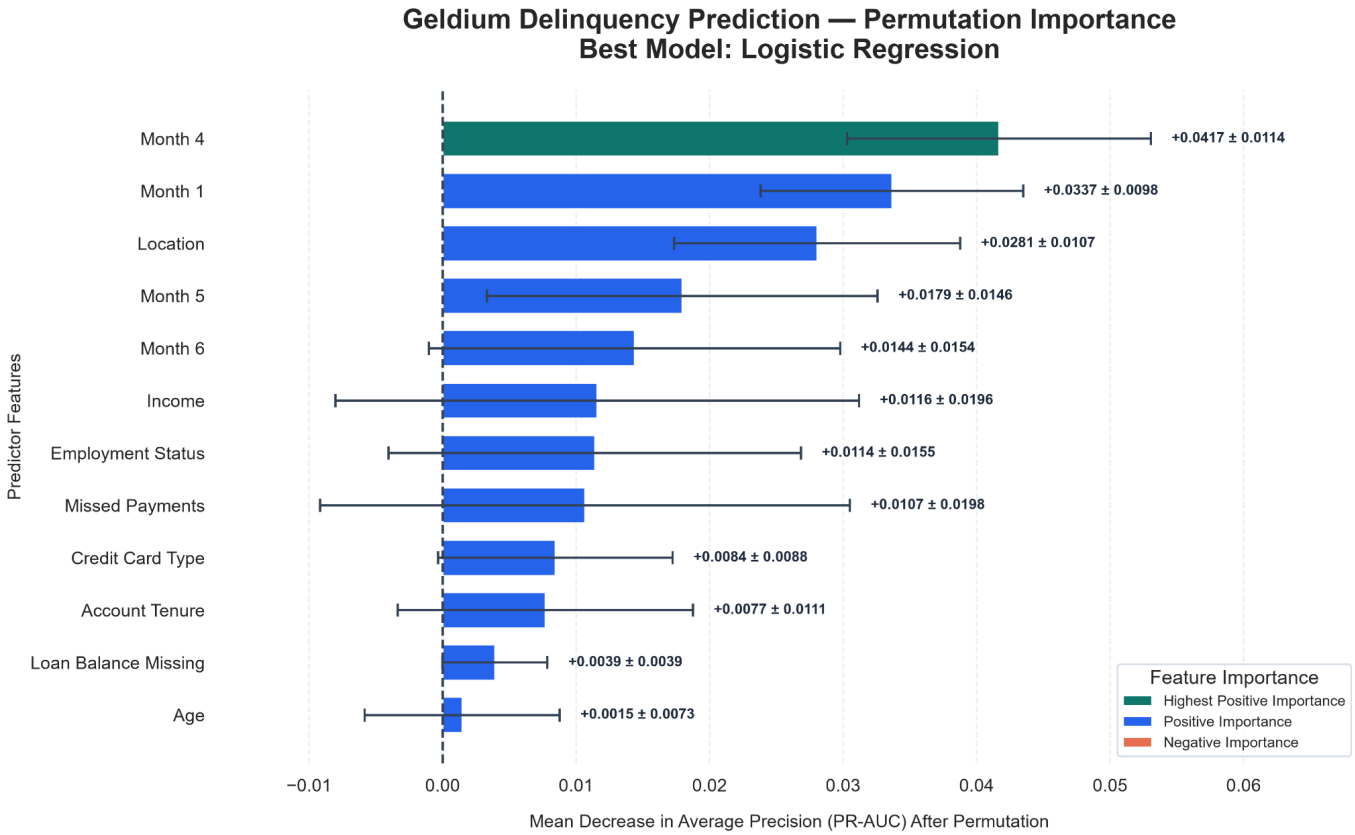


Figure 9. Change in holdout average precision after permutation of each original feature; bars show variability across permutations.

Month 4, Month 1 and Location show the largest displayed positive mean AP decreases (approximately 0.0417, 0.0337 and 0.0281). These are model-reliance measures, not causal effects. Overlapping error bars, correlated predictors, the small test set, and poor overall discrimination limit interpretation. In particular, Location can proxy for sensitive attributes, and monthly payment variables require a strict leakage check. Permutation importance on the final holdout should remain a descriptive audit, not a basis for repeated feature selection.

11 Exploratory fairness and subgroup review

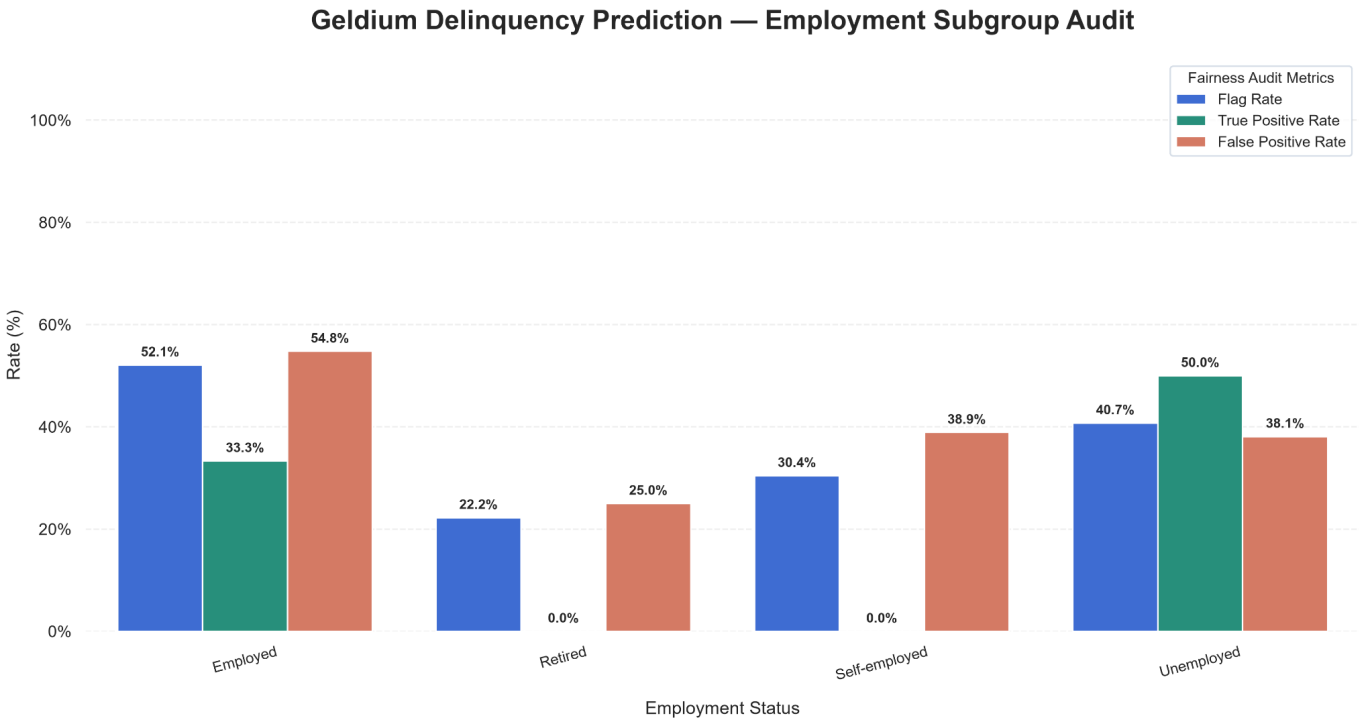


Figure 10. Logistic Regression flag rate, true-positive rate and false-positive rate by employment subgroup.

The chart shows substantial variation in flag rates and error rates: employed customers have a 52.1% flag rate and 54.8% false-positive rate, while retired and self-employed groups have zero observed true-positive rate in this holdout. Unemployed customers have a 50.0% true-positive rate. These rates cannot be interpreted without group denominators and positive-event counts; zero true-positive rate may result from only a handful of positives. Employment status is not equivalent to a legally protected-class audit, and it may itself be relevant to underwriting or a proxy for other attributes.

Minimum fairness audit before any operational use

Document sample size per group; compute group-wise precision, recall, FPR and selection rate with uncertainty intervals; compare equal-opportunity and equalized-odds differences where legally appropriate; test location and other proxy variables; review missing-data patterns and label bias; repeat across time and intersectional subgroups where sample sizes permit. Do not make regulatory compliance claims from this chart alone.

12 Model-by-model interpretation

Model	Empirical interpretation	Current role
Logistic Regression	Highest cross-validation AP (0.208); 5/20 positives detected with 44 false alarms; poor Brier 0.264.	Interpretable research benchmark; inspect leakage and calibration.
Decision Tree	Highest recall at 0.50 (45%) but 61 false positives; AP 0.146.	Illustrates sensitivity–specificity trade-off; unstable candidate.
Random Forest	Highest holdout AP in CSV (0.244), but ROC-AUC 0.400 and zero positives flagged at 0.50.	Exploratory challenger; no evidence of operational readiness.
Gradient Boosting	Holdout AP 0.186 and ROC-AUC 0.411; zero positives flagged.	No demonstrated incremental value.
Neural Network (MLP)	CV AP 0.206; holdout AP 0.182; zero true positives at 0.50.	Complexity not justified by observed performance.
Constant baseline	AP 0.160, ROC-AUC 0.500 and best Brier 0.134.	Mandatory benchmark for any future experiment.

Why the model family did not solve the problem

More complex learners cannot create reliable predictive information absent from the predictors. The EDA reported weak univariate associations; the holdout results suggest any multivariate patterns learned here do not generalize reliably. Before extensive tuning, verify label definitions, data provenance, chronology, representativeness and whether sufficient historical behavior exists to forecast a future event.

13 Remediation roadmap and deployment gates

Priority	Action	Evidence required
P0 — validity	Audit feature timestamps, delinquency label definition, leakage and duplicate/linked customers.	Documented point-in-time feature contract.
P1 — data	Acquire larger, time-stamped, representative history with enough delinquency events.	Coverage, drift, missingness and subgroup counts.
P1 — experimental	Repeat stratified/nested CV; use out-of-fold predictions for thresholds and calibration.	Confidence intervals and comparison to constant baseline.
P2 — features	Evaluate pre-event payment history, utilization trends, prior delinquency, account behavior.	Stable incremental lift under time-based validation.
P2 — operations	Define review capacity, intervention cost, harm, expected recovery and escalation.	Pre-registered decision threshold and cost/benefit analysis.
P3 — governance	Independent model validation, fairness review, adverse-action reason quality, monitoring and human oversight.	Approved model risk documentation and monitoring plan.

Explicit go / no-go requirements

A candidate should demonstrate repeatable out-of-time ranking improvement over the prevalence baseline; acceptable calibration for its intended use; stable precision and recall at a preselected workload; confidence intervals narrow enough to support a decision; defensible, reproducible explanations; and an approved fairness, privacy and model-governance review. Until then, use scores only for offline research—not customer denial, pricing, automated collections or adverse action.

GenAI use and analytical integrity

GenAI can help draft code, compare algorithm choices, structure documentation and translate verified diagnostics into plain-language narratives. It cannot validate unseen training runs, certify fairness, invent performance, or substitute for independent model review. All numerical findings in this report are based on the provided model-results CSV or visibly labeled figures; figure–CSV mismatches are disclosed rather than reconciled without evidence.

Conclusion

The main result is a defensible negative finding: the current Geldium experiment does not establish a reliable delinquency prediction model. The appropriate next investment is data/label validation and stronger prospective evidence—not deploying the model with the highest single holdout score.

Appendix A Metric definitions and audit notes

Metric	Formula / audit meaning
Precision	TP / (TP + FP): fraction of alerts that are correct.
Recall / TPR	TP / (TP + FN): fraction of delinquent customers detected.
FPR	FP / (FP + TN): fraction of non-delinquent customers incorrectly flagged.
F1	2 × precision × recall / (precision + recall).
Balanced accuracy	(TPR + TNR) / 2.
Brier score	Mean of (predicted probability − actual outcome)².
Cumulative gain at k%	Share of all observed delinquencies within the top k% ranked customers.
Lift at k%	Cumulative gain / k%.
Equal opportunity	Compare TPR across relevant groups, with sample-size uncertainty.
Equalized odds	Compare both TPR and FPR across relevant groups.

Source inventory

Primary numerical source: Geldium_Model_Results.csv. Supporting outputs: Geldium_Holdout_Risk_Scores.csv and the ten supplied PNG figures (01–10). Original data workbook: Delinquency_prediction_dataset.xlsx. The notebook and data should be version-controlled and re-executed in a frozen environment before external use.

Quality assurance note: Figure exports are preserved as provided; chart–CSV inconsistencies are explicitly disclosed in Section 1 and relevant captions. Model performance figures should not be combined across runs as if they came from one synchronized experiment.